

HDOC DAY-29

Title: Menu-Driven Array Operations: Counting, Searching, and Merging Arrays

1. Problem Statement Write a menu-driven C program to perform the following array operations:

1. Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge 2 arrays.
4. Exit.

2. Algorithms

1. Count Elements Algorithm

1. Read array size n and array elements $\text{arr}[n]$.
2. Initialize $\text{even}=0$, $\text{odd}=0$, $\text{positive}=0$, $\text{negative}=0$, $\text{zero}=0$.
3. Loop through array:
 - If $\text{arr}[i] \% 2 == 0$ increment even, else odd.
 - If $\text{arr}[i] > 0$ increment positive, else if $\text{arr}[i] < 0$ increment negative, else zero.

4. Output calculated totals.

2. Linear Search Algorithm

1. Read size n , array elements, and search key.
2. Iterate $i = 0$ to $n - 1$: if $\text{arr}[i] == \text{key}$, print index i and break.
3. If loop finishes without match, display element not found.

3. Merge Two Arrays Algorithm

1. Read sizes n_1 and n_2 and populate arr_1 and arr_2 .
2. Create $\text{merged}[n_1 + n_2]$.
3. Copy elements of arr_1 into $\text{merged}[0 \dots n_1 - 1]$ and arr_2 into $\text{merged}[n_1 \dots n_1 + n_2 - 1]$.
4. Print merged array.

3. Test Case Table

Test Case	Option	Input Data	Expected Output	Actual Output	Status
1	Count Elements	$N = 5$, elements: [-2, 0, 3, 4, -1]	Even: 3, Odd: 2, Pos: 2, Neg: 2, Zero: 1	Matched expected counts	Pass

Test Case	Option	Input Data	Expected Output	Actual Output	Status
2	Linear Search	$N = 4$, [10, 20, 30, 40], Key: 30	Found at index 2 (position 3)	Found at index 2	Pass
3	Linear Search	$N = 3$, [5, 15, 25], Key: 99	Element 99 not found	Element not found	Pass
4	Merge Arrays	A1: [1,2], , A2: [3,4,5]	Merged: [1, 2, 3, 4, 5]	Matched merged output	Pass
5	Exit	Option 4	Exiting program ...	Program exited	Pass

4. Source Code Listing

```
#include <stdio.h>
```

```

void count_array_elements(void) {
    int n;
    printf("\n--- 1. Count Elements ---\n");
    printf("Enter number of elements in array: ");
    scanf("%d", &n);
    if (n <= 0) return;

    int arr[n];
    int even = 0, odd = 0, positive = 0, negative = 0, zero = 0;

    printf("Enter %d integers:\n", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        if (arr[i] % 2 == 0) even++; else odd++;
        if (arr[i] > 0) positive++;
        else if (arr[i] < 0) negative++;
        else zero++;
    }

    printf("Even: %d, Odd: %d, Positive: %d, Negative: %d, Zeroes: %d\n",
        even, odd, positive, negative, zero);
}

void linear_search_array(void) {
    int n, key, found = 0;
    printf("\n--- 2. Linear Search ---\n");
    printf("Enter number of elements: ");
    scanf("%d", &n);
    if (n <= 0) return;

    int arr[n];
    printf("Enter elements:\n");
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);

    printf("Enter key: ");
    scanf("%d", &key);

    for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            printf("Element %d found at index %d.\n", key, i);

```

```

        found = 1;
        break;
    }
}
if (!found) printf("Element %d not found.\n", key);
}

void merge_two_arrays(void) {
    int n1, n2;
    printf("\n--- 3. Merge Two Arrays ---\n");
    printf("Enter size of first array: "); scanf("%d", &n1);
    int arr1[n1];
    for (int i = 0; i < n1; i++) scanf("%d", &arr1[i]);

    printf("Enter size of second array: "); scanf("%d", &n2);
    int arr2[n2];
    for (int i = 0; i < n2; i++) scanf("%d", &arr2[i]);

    int merged[n1 + n2];
    for (int i = 0; i < n1; i++) merged[i] = arr1[i];
    for (int i = 0; i < n2; i++) merged[n1 + i] = arr2[i];

    printf("Merged Array: [ ");
    for (int i = 0; i < n1 + n2; i++) printf("%d ", merged[i]);
    printf("]\n");
}

int main(void) {
    int choice;
    do {
        printf("\n1. Count Elements\n2. Linear Search\n3. Merge Arrays\n4.
Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: count_array_elements(); break;
            case 2: linear_search_array(); break;
            case 3: merge_two_arrays(); break;
            case 4: printf("Exiting program...\n"); break;
            default: printf("Invalid choice!\n"); break;
        }
    }
}

```

```
} while (choice != 4);  
return 0;  
}
```

5. Execution & Verification Screenshots

```
kali@kali:~/HMOOC_Day29  
Session Actions Edit View Help  
--(kali@kali)~  
$ mkdir HMOOC_Day29  
$ cd HMOOC_Day29  
--(kali@kali)~/HMOOC_Day29  
$ nano day29.c  
--(kali@kali)~/HMOOC_Day29  
$ gcc day29.c -o day29  
--(kali@kali)~/HMOOC_Day29  
$ ./day29  
  
===== ARRAY OPERATIONS MENU =====  
1. Count even, odd, positive, negative, and zero elements  
2. Search for an element in an array using linear search  
3. Merge 2 arrays  
4. Exit  
Enter choice (1-4): 1  
  
--- 1. Count Elements ---  
Enter number of elements in array: 5  
Enter 5 integers:  
Element 0: -2  
Element 1: 0  
Element 2: 3  
Element 3: 4  
Element 4: -1  
  
--- Element Counting Summary ---  
Even numbers: 3  
Odd numbers: 2  
Positive numbers: 2  
Negative numbers: 2  
Zeros: 1  
  
===== ARRAY OPERATIONS MENU =====  
1. Count even, odd, positive, negative, and zero elements  
2. Search for an element in an array using linear search  
3. Merge 2 arrays  
4. Exit  
Enter choice (1-4): 2  
  
--- 2. Linear Search ---  
Enter number of elements in array: 4  
Enter 4 integers:  
Element 0: 10  
Element 1: 20  
Element 2: 30  
Element 3: 40  
Enter element to search for: 30  
Element 30 found at index 2 (position 3).  
  
===== ARRAY OPERATIONS MENU =====  
1. Count even, odd, positive, negative, and zero elements  
2. Search for an element in an array using linear search  
3. Merge 2 arrays  
4. Exit  
Enter choice (1-4): 3  
  
--- 3. Merge Two Arrays ---  
Enter size of first array: 2  
Enter 2 elements for first array:  
1  
2  
Enter size of second array: 3  
Enter 3 elements for second array:  
3  
4  
5  
  
Merged Array: [ 1 2 3 4 5 ]  
  
===== ARRAY OPERATIONS MENU =====  
1. Count even, odd, positive, negative, and zero elements  
2. Search for an element in an array using linear search  
3. Merge 2 arrays  
4. Exit  
Enter choice (1-4): 4  
Exiting program...  
--(kali@kali)~/HMOOC_Day29
```